All phase-1 parts: what to print, what to keep

26 part files, 57 pieces. The 19 marked PRINT make the reprint set; the 7 marked KEEP are unchanged from the first print.

19 files
to print

42 pieces
on the bed

879 g
of filament

25h 09m
of printing

Filament to have ready (reprint set)

PLA Light Grey	516 g (buy with 20 % margin: 619 g)
PETG Orange	163 g (buy with 20 % margin: 195 g)
PLA Cobalt Blue	154 g (buy with 20 % margin: 185 g)
PLA Orange	46 g (buy with 20 % margin: 55 g)

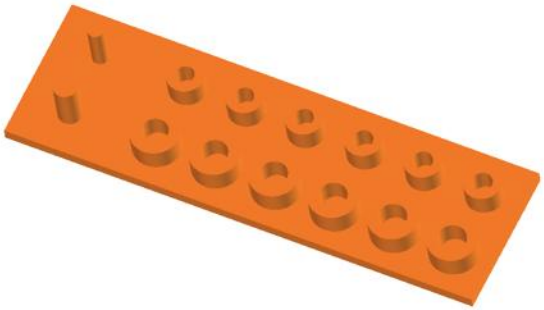
Keep from the first print

fit_coupon_x1: use your printed fit_coupon_x1
wheel_x6: use your printed wheel_x6
wedge_x1: use your printed wedge_x2 (you have 2, need 1)
lead_rail_230_x3: use your printed lead_rail_230_x3
chain_11_links_x3: use your printed chain_11_links_x3
gate_key_x1: use your printed gate_key_x4 (you have 4, need 1)
gate_pin_x1: use your printed gate_pin_x3 (you have 3, need 1)

Print order

1. Set 1 first: wagon parts (plates, panniers, pins, about 4.7 h), then the points and the handover.
2. Build one wagon with your existing wheels and roll it on an existing lead rail; work the points by hand.
3. Set 2 when set 1 works: combs, chain beds, sleds, bumpers, window hook, start gate, lever.

Parts 1 to 3 of 26



KEEP from the first print Test piece

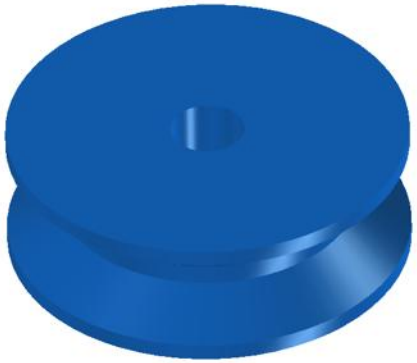
fit_coupon_x1

PETG Orange

qty 1 · 21.5 g · 0h 59m · 118.0 × 36.0 × 15.0 mm on the bed · 0.12 mm layers

print first: posts and bored rings at 0.15 to 0.40 mm per side

Use your printed fit_coupon_x1.



KEEP from the first print Wagon

wheel_x6

PLA Cobalt Blue

qty 6 · 20.9 g · 1h 06m · 24.0 × 24.0 × 8.0 mm on the bed · 0.12 mm layers

V-groove wheel; rides the wire and the rail head

Use your printed wheel_x6.



PRINT Wagon

axle_pin_x6

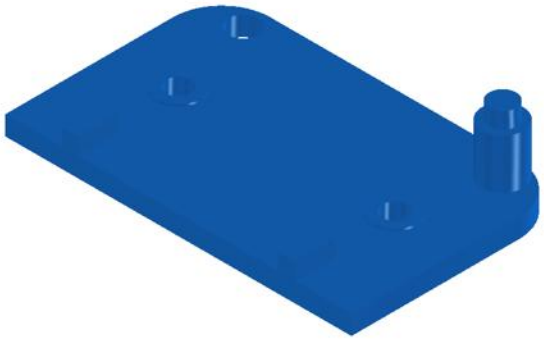
PETG Orange

qty 6 · 3.4 g · 0h 52m · 8.0 × 8.0 × 32.4 mm on the bed · 0.12 mm layers

4 mm pin through pannier, plate, wheel, plate, pannier; snap bump

Why reprint: longer (28.4 mm shank) to pass through pannier, plate, wheel, plate, pannier; snap bump 0.7 mm clear

Parts 4 to 6 of 26



PRINT Wagon

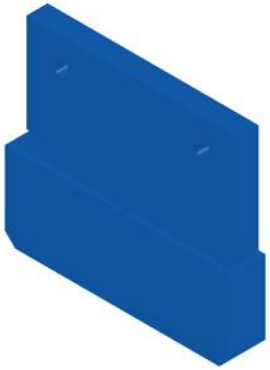
trolley_plate_x6

PLA Cobalt Blue

qty 6 · 49.1 g · 1h 27m · 64.0 × 40.0 × 15.8 mm on the bed · 0.20 mm layers

two identical plates per wagon; prints flat, peg up

Why reprint: pegs reach the other plate and carry shoulders that set the 9.2 mm gap; axle holes print round (the plate lies flat); lips 0.5 mm clear of the web; 3.6 thick with 0.4 bosses to suit 0.2 mm layers



PRINT Wagon

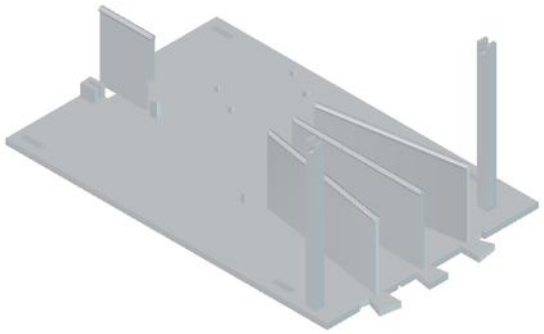
pannier_x6

PLA Cobalt Blue

qty 6 · 105.2 g · 2h 25m · 64.0 × 10.2 × 54.8 mm on the bed · 0.20 mm layers

basket; two identical per wagon, clamped by the axle pins

Why reprint: new: one part for both sides, clamped by the axle pins, floor 12 mm above the strip; arm spacing for the 9.2 mm plate gap



PRINT Points

turnout_base_x1

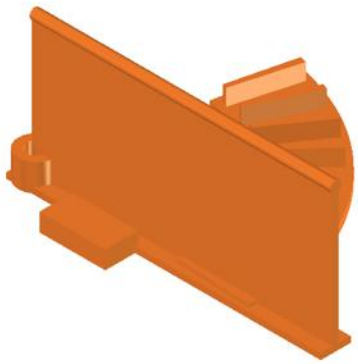
PLA Light Grey

qty 1 · 159.9 g · 3h 38m · 249.0 × 155.0 × 109.0 mm on the bed · 0.20 mm layers

plate, entry and exit rails, pivot, stops, rocker bridges, gantry posts

Why reprint: new layout: lane spacing, pegs for the rocker bridge, gantry posts with 3.6 mm snap mouths, wire groove underneath; the entry rail's end lets the tongue slide under

Parts 7 to 9 of 26



PRINT Points

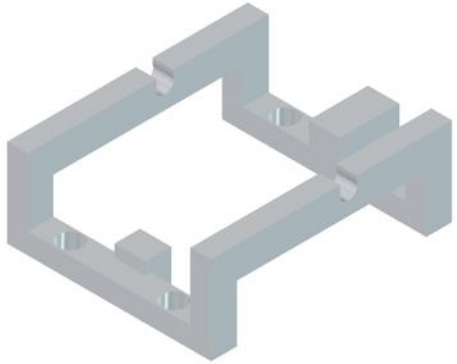
tongue_x1

PLA Orange

qty 1 · 22.5 g · 0h 41m · 106.0 × 88.7 × 49.5 mm on the bed · 0.20 mm layers

switch blade with the ring of four teeth

Why reprint: new: ring of four teeth and the spring tab



PRINT Points

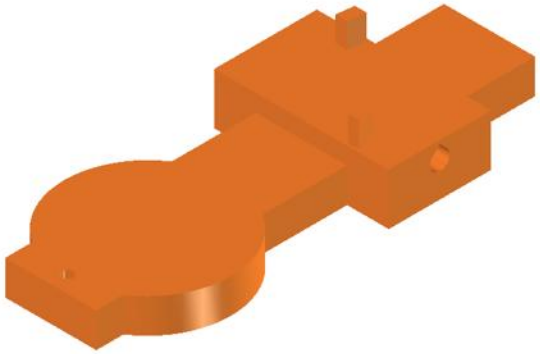
rocker_bridge_x1

PLA Light Grey

qty 1 · 6.6 g · 0h 25m · 39.4 × 53.0 × 17.0 mm on the bed · 0.20 mm layers

two beams over the tooth ring; carries the rocker pin and the tilt stops; sits on four pegs

Why reprint: new: carries the rocker over the tooth ring; separate so the tongue can be fitted first; beams 2.5 mm over the teeth, sliced with supports



PRINT Points

rocker_x1

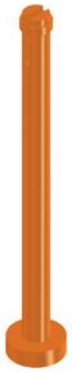
PLA Orange

qty 1 · 8.3 g · 0h 28m · 30.6 × 77.0 × 13.4 mm on the bed · 0.12 mm layers

escapement rocker; prints upside down

Why reprint: new: escapement rocker with a 26 mm coin cup

Parts 10 to 12 of 26



PRINT Points

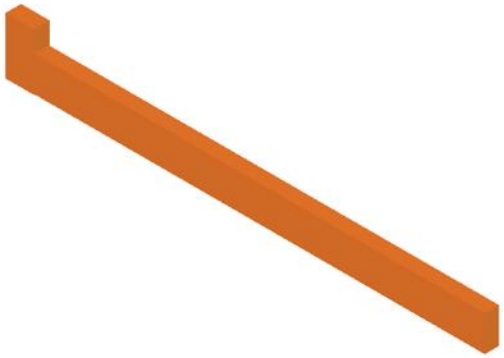
rocker_pin_x1

PETG Orange

qty 1 · 0.5 g · 0h 14m · 6.5 × 6.5 × 44.0 mm on the bed · 0.12 mm layers

3 mm rocker axle with snap bump

Why reprint: new



PRINT Points

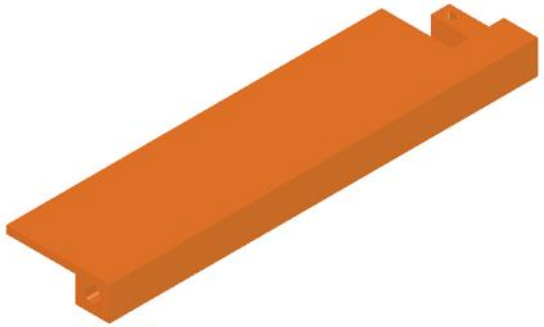
bias_spring_x1

PETG Orange

qty 1 · 1.0 g · 0h 14m · 67.4 × 2.2 × 8.0 mm on the bed · 0.12 mm layers

leaf spring that pushes the tongue toward the next lane

Why reprint: new leaf spring



PRINT Points

flap_x1

PLA Orange

qty 1 · 14.8 g · 0h 25m · 34.5 × 114.0 × 9.0 mm on the bed · 0.20 mm layers

trigger flap with crank; prints on its downstream face

Why reprint: new trigger flap with a 16 mm crank

Parts 13 to 15 of 26



PRINT Points

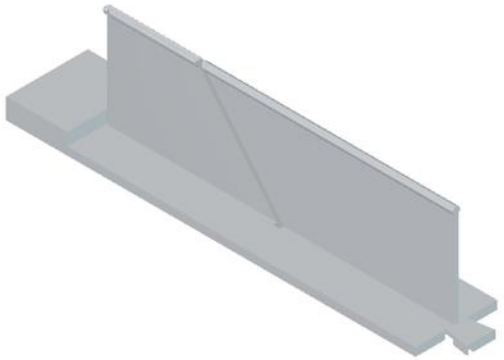
gantry_axle_x1

PETG Orange

qty 1 · 2.1 g · 0h 12m · 4.0 × 134.0 × 3.5 mm on the bed · 0.12 mm layers

4 mm flap axle, prints on its flat

Why reprint: new



PRINT Wire end

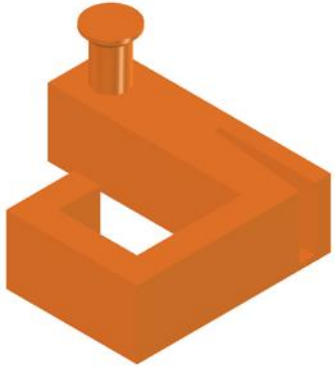
handover_x1

PLA Light Grey

qty 1 · 49.1 g · 1h 06m · 212.0 × 40.0 × 53.5 mm on the bed · 0.20 mm layers

wire-to-rail handover; the wire dives through it

Why reprint: the wire now dives through it and runs on under the yard; head height matches the wheel seat



PRINT Wire end

window_hook_x1

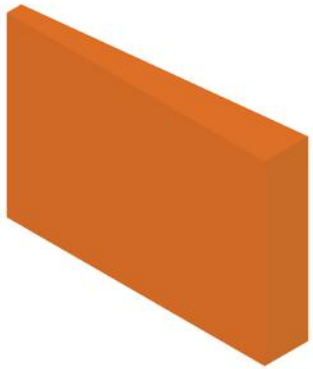
PETG Orange

qty 1 · 49.6 g · 1h 25m · 84.0 × 74.0 × 46.0 mm on the bed · 0.20 mm layers

top anchor: capstan post and wedge pocket

Why reprint: pocket opens upward; rim prints without support

Parts 16 to 18 of 26



KEEP from the first print Wire end

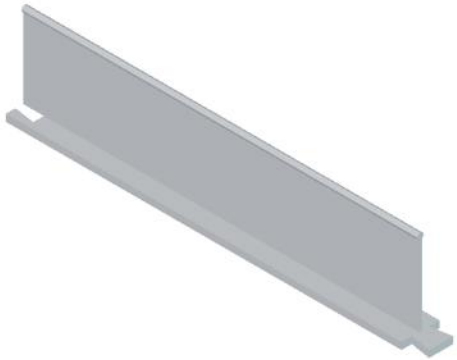
wedge_x1

PETG Orange

qty 1 · 3.2 g · 0h 39m · 32.0 × 5.4 × 22.0 mm on the bed · 0.12 mm layers

pinches the wire in the window hook

Use your printed wedge_x2 (you have 2, need 1).



KEEP from the first print Bays

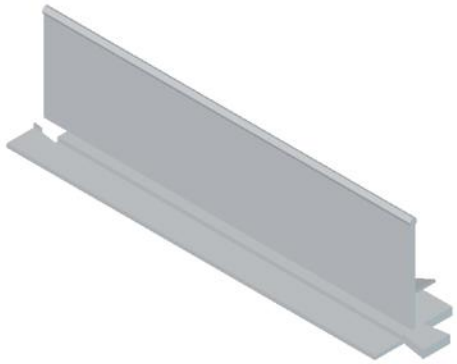
lead_rail_230_x3

PLA Light Grey

qty 3 · 131.5 g · 2h 29m · 242.0 × 22.0 × 53.5 mm on the bed · 0.20 mm layers

first bay segment, narrow strip

Use your printed lead_rail_230_x3.



PRINT Bays

brake_comb_190_x3

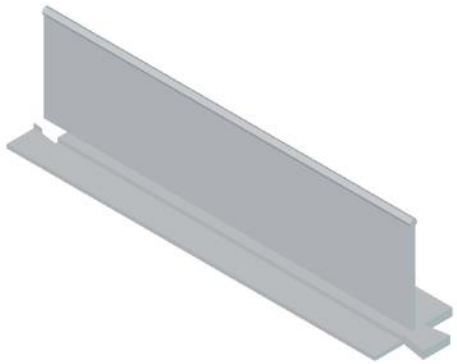
PLA Light Grey

qty 3 · 150.5 g · 3h 27m · 202.0 × 40.0 × 53.5 mm on the bed · 0.20 mm layers

rail with 12 leaves under the right basket and the chain trough under the left

Why reprint: v1 leaves stood 21.7 mm tall and would ram the baskets; v2 leaves reach 13.8 mm and end inside the segment

Parts 19 to 21 of 26



PRINT Bays

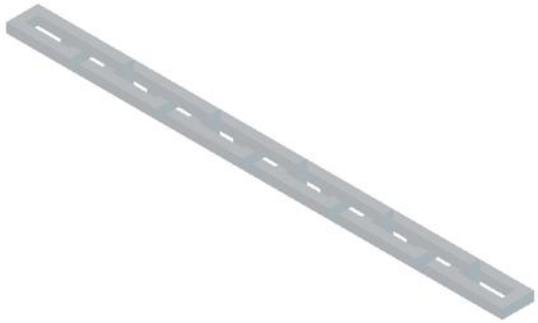
chain_bed_190_x3

PLA Light Grey

qty 3 · 132.9 g · 2h 34m · 202.0 × 40.0 × 53.5 mm on the bed · 0.20 mm layers

rail with the chain trough, end to end; the sled starts here

Why reprint: trough runs to the end so the sled can reach the bumper; no stirrup ring



KEEP from the first print Bays

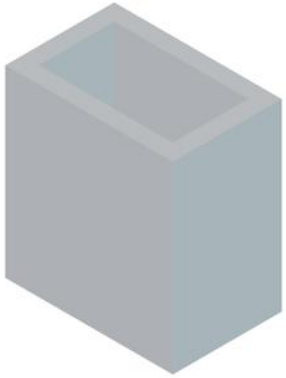
chain_11_links_x3

PLA Light Grey

qty 3 · 17.1 g · 0h 59m · 156.0 × 10.0 × 10.0 mm on the bed · 0.12 mm layers

print-in-place drag chain

Use your printed chain_11_links_x3.



PRINT Bays

sled_x3

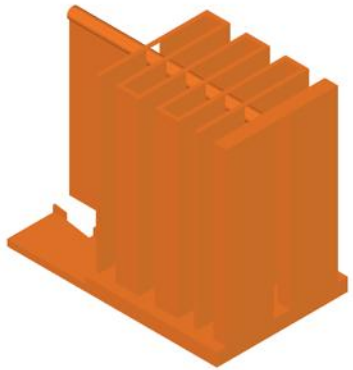
PLA Light Grey

qty 3 · 10.0 g · 0h 23m · 20.0 × 13.0 × 22.0 mm on the bed · 0.20 mm layers

pushed by the left basket; drags the chain tied to its back

Why reprint: new: the left basket pushes it and the chain

Parts 22 to 24 of 26



PRINT Bays

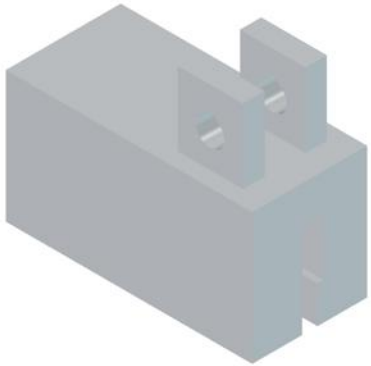
bumper_x3

PETG Orange

qty 3 · 106.2 g · 4h 45m · 64.0 × 40.0 × 56.0 mm on the bed · 0.12 mm layers

zig-zag spring at the end of each bay

Why reprint: rail stub runs to the face plate; trough for the sled



PRINT Start gate

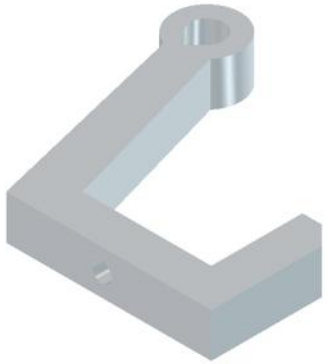
start_gate_x1

PLA Light Grey

qty 1 · 5.8 g · 0h 19m · 30.0 × 14.0 × 25.0 mm on the bed · 0.20 mm layers

clamps on the wire behind the waiting wagon

Why reprint: v1 wire slot was 1 mm for a 1.5 mm wire and the key missed the wire; the slot now opens downward so the gate hangs on the wire



PRINT Start gate

gate_lever_x1

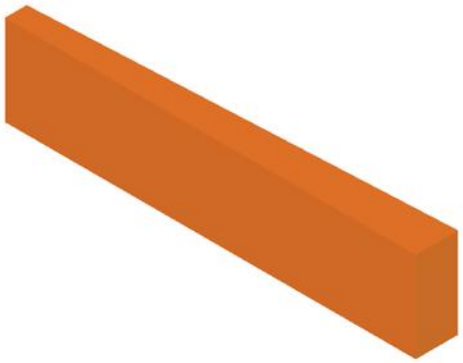
PLA Light Grey

qty 1 · 1.4 g · 0h 09m · 28.9 × 28.5 × 4.6 mm on the bed · 0.20 mm layers

hook over the wagon's rear spacer peg

Why reprint: new hook over the rear spacer peg

Parts 25 to 26 of 26



KEEP from the first print Start gate

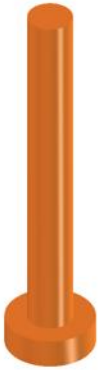
gate_key_x1

PETG Orange

qty 1 · 0.7 g · 0h 14m · 30.0 × 3.1 × 6.6 mm on the bed · 0.12 mm layers

tapered key that locks the gate on the wire

Use your printed gate_key_x4 (you have 4, need 1).



KEEP from the first print Start gate

gate_pin_x1

PETG Orange

qty 1 · 0.4 g · 0h 11m · 6.0 × 6.0 × 26.0 mm on the bed · 0.12 mm layers

lever pivot pin

Use your printed gate_pin_x3 (you have 3, need 1).